28 Self-Play and AlphaZero

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Monte Carlo tree search in base R, the self-play training paradigm, and how AlphaZero combines MCTS with deep learning to master board games from scratch.

Learning objectives

- Explain why self-play is a powerful training paradigm for two-player zero-sum games.
- Implement Monte Carlo Tree Search (MCTS) in base R, including UCT selection, rollout, and backpropagation.
- Describe the AlphaZero architecture and how neural networks replace the rollout phase of MCTS.
- Inspect MCTS visit counts and measure finite-budget play against a random opponent.

28.1 Motivation

In 26 and 27, we trained agents using tabular methods on small matrix games. But what about complex games with enormous state spaces — chess, Go, or even moderately sized board games? Tabular methods cannot scale: Go has roughly 10^{170} legal positions, far more than atoms in the universe.

Self-play offers an elegant solution. Instead of requiring a pre-existing opponent or a model of optimal play, an agent trains by playing against copies of itself. Each generation of the agent becomes a training partner for the next. This bootstrapping process, combined with Monte Carlo Tree Search (MCTS), enabled AlphaGo to defeat the world Go champion (Silver et al., 2016) and AlphaZero to master chess, shogi, and Go from scratch with no human knowledge beyond the rules (Silver et al., 2017).

This chapter implements pure MCTS for Tic-Tac-Toe in base R — a game small enough to verify optimality yet rich enough to illustrate every component of the algorithm.

28.2 Theory

28.2.1 Self-play as a training methodology

In a two-player zero-sum game, any improvement in one player's strategy creates a harder training signal for the other. Self-play exploits this: initialize a policy, play it against itself, use outcomes to improve, and repeat. The key property is **automatic curriculum generation** — as the agent improves, its opponent improves in lockstep, continuously providing appropriately challenging games without hand-crafted opponents.

28.2.2 Monte Carlo Tree Search

MCTS builds a search tree incrementally through four repeated phases:

1. **Selection**: Starting from the root (current game state), traverse the tree by choosing child nodes according to a **tree policy** until reaching a leaf node. The standard tree policy is **Upper Confidence Bound for Trees (UCT)**:

$$\mathrm{UCT}(s,a) = \bar{Q}(s,a) + c\sqrt{\frac{\ln N(s)}{N(s,a)}} \tag{28.1}$$

where $\bar{Q}(s,a)$ is the average outcome of action a from state s, $N(s)$ is the visit count of state s, $N(s,a)$ is the visit count of action a from s, and c is an exploration constant (typically $\sqrt{2}$).

2. **Expansion**: Add one or more child nodes to the tree at the selected leaf.

3. **Rollout** (simulation): From the newly expanded node, play a complete game using a **default policy** (typically random moves) to obtain an outcome.

4. **Backpropagation**: Update the visit counts and value estimates along the path from the expanded node back to the root.

After a fixed number of iterations, the algorithm selects the action from the root with the highest visit count — not the highest average value, because visit count is a more robust estimator.

28.2.3 AlphaZero architecture

AlphaZero (Silver et al., 2017) replaces the random rollout in step 3 with a **neural network** that directly evaluates positions. The network $f_\theta(s) = (\mathbf{p}, v)$ takes a board state s and outputs:

- A **policy vector** $\mathbf{p}$: prior probabilities over legal moves, used in a prior-weighted tree policy (PUCT).
- A **value estimate** $v \in [-1, 1]$: the predicted game outcome from state s.

The training loop iterates: play self-play games using MCTS guided by f_θ, store (s_t, π_t, z_t) tuples (state, MCTS visit-count policy, outcome), and train f_θ to minimize $(v - z)^2 - \pi^\top \log \mathbf{p} + c\|\theta\|^2$ (value error, policy cross-entropy, and weight regularization). AlphaZero learned chess at superhuman level using only the rules (Silver et al., 2017).

Since neural networks require deep learning frameworks beyond our available packages, we implement **pure MCTS** (with random rollouts) below. This illustrates tree search, not AlphaZero training: AlphaZero also uses policy priors in selection, learns a value function, and updates a network from self-play.

28.3 Implementation in R

28.3.1 Tic-Tac-Toe game engine

```
ttt_new ← function() rep(0L, 9)  # 0 = empty, 1 = X, -1 = O

ttt_moves ← function(board) which(board == 0L)

ttt_play ← function(board, pos, player) {
  if (length(pos) ≠ 1L || !is.numeric(pos) || !is.finite(pos) ||
      pos ≠ trunc(pos) || !pos %in% ttt_moves(board))
    stop("pos must identify one empty square")
  if (length(player) ≠ 1L || is.na(player) ||
      player ≠ ttt_current_player(board) || !is.na(ttt_winner(board)))
    stop("player must be the side to move in an unfinished game")
  board[pos] ← player
  board
}

ttt_winner ← function(board) {
  lines ← list(1:3, 4:6, 7:9,           # rows
               c(1,4,7), c(2,5,8), c(3,6,9),  # cols
               c(1,5,9), c(3,5,7))        # diagonals
  for (l in lines) {
    s ← sum(board[l])
    if (s ==  3) return(1L)   # X wins
    if (s == -3) return(-1L)  # O wins
  }
  if (all(board ≠ 0L)) return(0L)  # draw
  NA_integer_  # game ongoing
}

ttt_current_player ← function(board) {
  if (sum(board == 1L) == sum(board == -1L)) 1L else -1L
}
```

28.3.2 MCTS implementation

```
mcts_node ← function(board, parent = NULL, action = NULL) {
  # Environments give parent/child references shared mutable state.
  list2env(list(
    board = board,
    parent = parent,
    action = action,
    children = list(),
    visits = 0,
    total_value = 0,
    untried = if (is.na(ttt_winner(board))) ttt_moves(board) else integer()
  ), parent = emptyenv())
}

mcts_select ← function(node, c_explore = sqrt(2)) {
  # UCT selection: descend tree until we find a node with untried moves or terminal
  while (length(node$untried) == 0 && length(node$children) > 0) {
    player ← ttt_current_player(node$board)
    best_val ← -Inf
    best_child ← NULL
    for (child in node$children) {
      q ← child$total_value / child$visits
      # Flip sign for opponent's perspective
      uct ← player * q + c_explore * sqrt(log(node$visits) / child$visits)
      if (uct > best_val) {
        best_val ← uct
        best_child ← child
      }
    }
    node ← best_child
  }
  node
}

mcts_expand ← function(node) {
  if (length(node$untried) == 0) return(node)
  action ← node$untried[sample.int(length(node$untried), 1L)]
  node$untried ← setdiff(node$untried, action)
  player ← ttt_current_player(node$board)
  new_board ← ttt_play(node$board, action, player)
  child ← mcts_node(new_board, parent = node, action = action)
  node$children ← c(node$children, list(child))
  child
}

mcts_rollout ← function(board) {
  while (is.na(ttt_winner(board))) {
    player ← ttt_current_player(board)
    moves ← ttt_moves(board)
    board ← ttt_play(board, moves[sample.int(length(moves), 1L)], player)
  }
  ttt_winner(board)
}

mcts_backpropagate ← function(node, result) {
  while (!is.null(node)) {
    node$visits ← node$visits + 1
    node$total_value ← node$total_value + result
    node ← node$parent
  }
}
```

28.3.3 Running MCTS

```
mcts_search ← function(board, n_iterations = 1000, c_explore = sqrt(2)) {
  if (!is.numeric(board) || length(board) ≠ 9L || anyNA(board) ||
      any(!board %in% c(-1L, 0L, 1L)) ||
      !sum(board) %in% c(0, 1)) stop("board must be a legal nine-square position")
  if (length(n_iterations) ≠ 1L || !is.numeric(n_iterations) ||
      !is.finite(n_iterations) || n_iterations < 1 ||
      n_iterations ≠ trunc(n_iterations) || n_iterations > .Machine$integer.max)
    stop("n_iterations must be a positive integer")
  if (length(c_explore) ≠ 1L || !is.numeric(c_explore) ||
      !is.finite(c_explore) || c_explore < 0) stop("c_explore must be finite and
nonnegative")
  if (!is.na(ttt_winner(board))) stop("search requires an unfinished game")
  root ← mcts_node(board)

  for (i in seq_len(n_iterations)) {
    # Selection
    node ← mcts_select(root, c_explore)
    # Check if terminal
    winner ← ttt_winner(node$board)
    if (is.na(winner)) {
      # Expansion
      node ← mcts_expand(node)
      # Rollout
      result ← mcts_rollout(node$board)
    } else {
      result ← winner
    }
    # Backpropagation
    mcts_backpropagate(node, result)
  }

  # Return root node info
  move_stats ← tibble(
    move = as.integer(sapply(root$children, function(ch) ch$action)),
    visits = as.integer(sapply(root$children, function(ch) ch$visits)),
    avg_value = as.double(sapply(root$children, function(ch) ch$total_value /
max(ch$visits, 1)))
  ) ▷
    arrange(desc(visits))

  list(root = root, stats = move_stats,
       best_move = move_stats$move[1])
}
```

28.3.4 MCTS tree visualization

```
set.seed(42)
result ← mcts_search(ttt_new(), n_iterations = 100)

# Build a data frame for the board grid
stats ← result$stats
board_df ← tibble(
  pos = 1:9,
  row = rep(3:1, each = 3),
  col = rep(1:3, 3)
) ▷
  left_join(
    stats ▷ rename(pos = move),
    by = "pos"
  ) ▷
  mutate(
    visits = replace_na(visits, 0),
    avg_value = replace_na(avg_value, 0),
    label = if_else(visits > 0,
                    paste0("V:", visits, "\nS:", sprintf("%.0f%%",
                           (avg_value + 1) / 2 * 100)),
                    "")
  )

p_tree ← ggplot(board_df, aes(x = col, y = row)) +
  geom_tile(aes(fill = visits), colour = "grey30", linewidth = 1.2) +
  geom_text(aes(label = label), size = 3.5, fontface = "bold") +
  scale_fill_gradient(low = "white", high = okabe_ito[2],
                      name = "Visit Count") +
  theme_publication() +
  theme(
    axis.text = element_blank(),
    axis.ticks = element_blank(),
    panel.grid = element_blank(),
    aspect.ratio = 1
  ) +
  labs(title = "MCTS Visit Counts from Empty Board",
       subtitle = "100 iterations, X to move",
       x = NULL, y = NULL)

p_tree
```

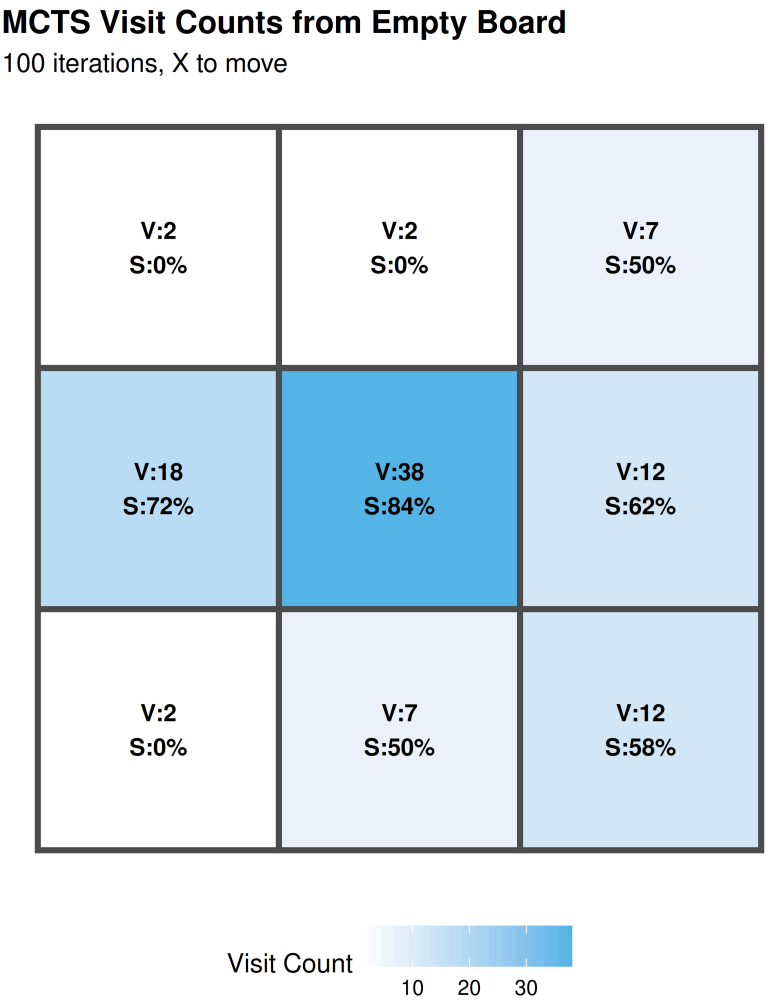


Figure 28.1: MCTS search from an empty Tic-Tac-Toe board (1000 iterations). Each cell shows visits and mean score from X's perspective (win = 1, draw = 0.5, loss = 0). Finite-budget visit counts reflect this seeded search, not a proof that an opening is uniquely optimal.

```
save_pub_fig(p_tree, "mcts-tree-visits", width = 6, height = 6)
```

28.3.5 Evaluation against a random opponent

This experiment does not train a policy or perform self-play. It measures X's results in 50 games per search budget against a random O; sampling uncertainty and the first-player advantage matter.

```
mcts_vs_random ← function(n_iterations) {
  board ← ttt_new()
  while (is.na(ttt_winner(board))) {
    player ← ttt_current_player(board)
    if (player == 1L) {
      # MCTS player
      res ← mcts_search(board, n_iterations = n_iterations)
      board ← ttt_play(board, res$best_move, player)
    } else {
      # Random opponent
      moves ← ttt_moves(board)
      board ← ttt_play(board, moves[sample.int(length(moves), 1L)], player)
    }
  }
  ttt_winner(board)
}

set.seed(42)
iteration_levels ← c(10, 25, 50, 100, 200, 500)
n_games ← 50

self_play_results ← tibble(
  iterations = integer(),
  win_rate = double(),
  draw_rate = double()
)

for (n_iter in iteration_levels) {
  outcomes ← replicate(n_games, mcts_vs_random(n_iter))
  self_play_results ← bind_rows(self_play_results, tibble(
    iterations = n_iter,
    win_rate = mean(outcomes == 1),
    draw_rate = mean(outcomes == 0)
  ))
}
```

```
df_sp ← self_play_results ▷
  pivot_longer(c(win_rate, draw_rate), names_to = "outcome",
               values_to = "rate") ▷
  mutate(outcome = recode(outcome,
                          "win_rate" = "Win", "draw_rate" = "Draw"))

p_sp ← ggplot(df_sp, aes(x = iterations, y = rate, colour = outcome)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  scale_x_log10(breaks = iteration_levels) +
  scale_y_continuous(labels = scales::percent, limits = c(0, 1)) +
  scale_colour_manual(values = c("Win" = okabe_ito[3], "Draw" = okabe_ito[1])) +
  theme_publication() +
  labs(title = "MCTS Strength vs. Search Budget",
       x = "MCTS Iterations per Move (log scale)",
       y = "Rate against Random Opponent",
       colour = "Outcome")

p_sp
```

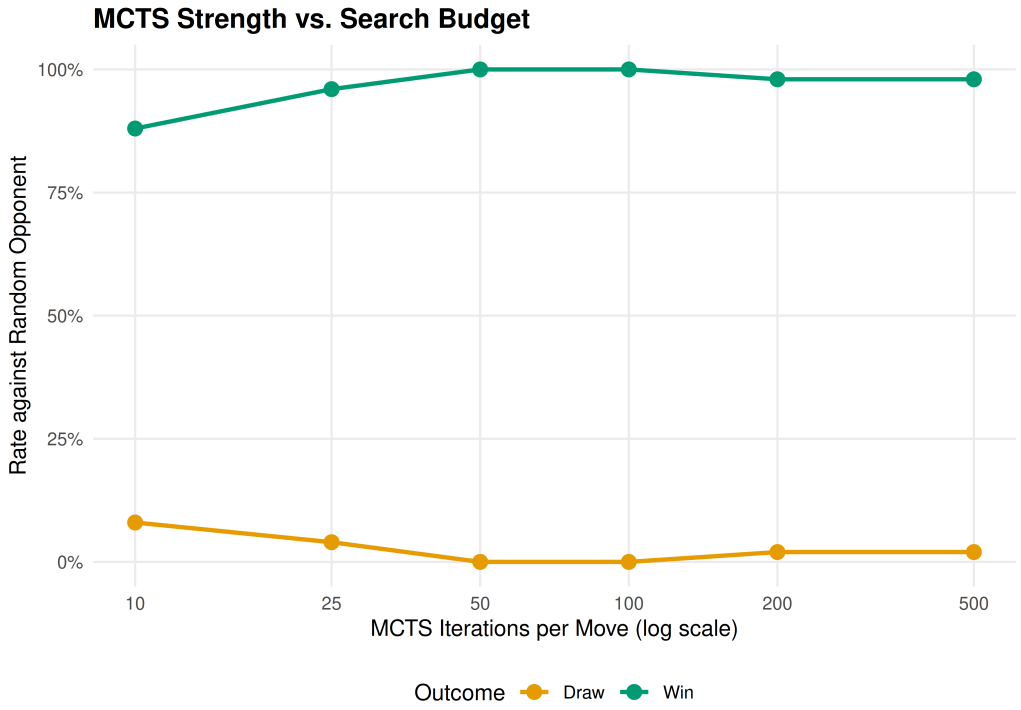


Figure 28.2: MCTS win rate against a random opponent as search budget increases. Each point summarizes 50 seeded games with MCTS playing X. These finite samples do not establish monotonic improvement or optimality.

```
save_pub_fig(p_sp, "mcts-self-play-winrate", width = 7, height = 5)
```

28.4 Worked example

We trace MCTS solving a mid-game Tic-Tac-Toe position step by step. Consider this board state where X has a forced win:

```
 X | X | .
-----------
 O | O | .
-----------
 . | . | .
```

X occupies positions 1 and 2; O occupies 4 and 5. It is X's turn, and position 3 wins immediately.

```
board_mid ← rep(0L, 9)
board_mid[1] ← 1L   # X top-left
board_mid[2] ← 1L   # X top-middle
board_mid[4] ← -1L  # O middle-left
board_mid[5] ← -1L  # O center

cat("Board state:\n")
```

```
#> Board state:
```

```
symbols <- c("-1" = "O", "0" = ".", "1" = "X")
for (r in 1:3) {
  row_str <- paste(symbols[as.character(board_mid[(r-1)*3 + 1:3])],
                   collapse = " | ")
  cat(" ", row_str, "\n")
  if (r < 3) cat("  ---------\n")
}
```

```
#>   X | X | .
#>   ---------
#>   O | O | .
#>   ---------
#>   . | . | .
```

```
set.seed(42)
result_mid <- mcts_search(board_mid, n_iterations = 2000)

cat("\nMCTS analysis (2000 iterations):\n")
```

```
#>
#> MCTS analysis (2000 iterations):
```

```
cat("Move | Visits | Mean Score (X)\n")
```

```
#> Move | Visits | Mean Score (X)
```

```
cat("-----+--------+-------------\n")
```

```
#> -----+--------+-------------
```

```
for (i in seq_len(nrow(result_mid$stats))) {
  s <- result_mid$stats[i, ]
  cat(sprintf("  %d  |  %4d  |   %.1f%%\n",
              s$move, s$visits, (s$avg_value + 1) / 2 * 100))
}
```

```
#>   3  |  1958  |   100.0%
#>   6  |    20  |   60.0%
#>   9  |     9  |   38.9%
#>   8  |     7  |   28.6%
#>   7  |     6  |   16.7%
```

```
cat(sprintf("\nBest move: position %d\n", result_mid$best_move))
```

```
#>
#> Best move: position 3
```

```
# Check the resulting board rather than assume a tactical explanation.
best ← result_mid$best_move
if (isTRUE(ttt_winner(ttt_play(board_mid, best, 1L)) == 1L)) {
  cat("The selected move completes X's top row and wins immediately.\n")
} else {
  cat("This finite search missed the immediate win at position 3.\n")
}
```

```
#> The selected move completes X's top row and wins immediately.
```

The visit count distribution reveals MCTS's reasoning: the distribution records how this finite search allocated its simulations. A winning move can be verified directly by the game rules. UCT ((28.1)) balances exploration and exploitation without domain knowledge — a principled solution to the multi-armed bandit problem applied recursively through the game tree.

28.5 Extensions

- **AlphaZero's neural MCTS**: Replace random rollouts with a neural network evaluation trained on self-play data, creating a virtuous cycle of improving evaluation, search, and training data (Silver et al., 2017).
- **Progressive widening**: In games with large branching factors, expand moves proportional to visit count, focusing computation on promising subtrees.
- **RAVE (Rapid Action Value Estimation)**: Share value estimates across sibling nodes to accelerate convergence in early search.
- **Temperature-based move selection**: Sample moves proportional to $N(s,a)^{1/\tau}$ during training to maintain exploration diversity.
- **Counterfactual regret minimization** (29) provides an alternative tree-search framework for imperfect-information games where MCTS struggles.
- **Multi-agent RL** (27) covers how to train agents in games with more than two players.

Exercises

1. **Exploration constant sensitivity.** Run MCTS on the empty Tic-Tac-Toe board with $c \in \{0.5, \sqrt{2}, 3.0\}$ for 500 iterations each. Plot the visit count distribution across the 9 opening moves for each value of c. How does the exploration constant affect the concentration of visits?

2. **MCTS vs. MCTS.** Implement a self-play match where both X and O use MCTS (with different random seeds). Run 100 games with 200 iterations per move for each player. What fraction are draws? In perfect play, Tic-Tac-Toe is always a draw — how close does MCTS get with this search budget?

3. **Game tree size.** Tic-Tac-Toe has at most $9! = 362{,}880$ possible game sequences (fewer accounting for terminal states). Write a function that exactly counts the number of distinct game sequences that end in X winning, O winning, and a draw using exhaustive enumeration (recursion). Compare the exact win probabilities under random play with the MCTS estimates from the worked example.

Solutions appear in D.